

JAKE JABS COLLEGE OF BUSINESS & ENTREPRENEURSHIP · MONTANA STATE UNIVERSITY-BOZEMAN

BMGT 240IS — Business Analytics

Fall 2026 Syllabus · Section 002 · CRN 23266

Term	2026 Fall Semester (08/27/2026 – 12/17/2026)
Credits	3
Meeting Times	Tuesday & Thursday, 12:15 – 1:30 pm
Location	JABS 215, MSU-Bozeman
Format	Face to Face — Lecture
Instructor	Xuying W. Swift
Email	xuying.swift@montana.edu
Office / Hours	<i>TBD — see Brightspace</i>
Grade Mode	Traditional letter grade
Course Site	D2L Brightspace

Course Description

The fundamental purpose of this course is to enable students to understand basic statistical concepts and techniques, and how they can be applied in the field of business by modern managers. The course aims to build your statistical background and to give you the statistical skills necessary to meet the needs of industry and real-world decision-making problems. Emphasis is on statistical reasoning, problem-solving, computer applications, and interpretation of results.

Learning Objectives

By the end of this course you will be able to:

1. Gather, clean, describe, and summarize data using Excel
2. Take samples and estimate unknown parameters
3. Formulate and test hypotheses
4. Understand basic principles of statistical process control
5. Use multiple regression to analyze data
6. Successfully complete research projects
7. Design and conduct research in the future
8. Use sound quantitative reasoning to analyze business problems

Textbook

David P. Doane and Lori E. Seward, *Applied Statistics in Business and Economics*, 7th edition — **recommended, not required**. All required material is presented in class and posted to Brightspace. Chapter references in the schedule below are provided for students who want the extra reading.

Technology Requirements

- **Microsoft Excel** (desktop version) with the **Analysis ToolPak** add-in enabled. Excel is free to MSU students through Microsoft 365; the browser version will *not* support all the tools we use.
- A laptop brought to class on lab days (marked **Lab** in the schedule).
- Reliable access to D2L Brightspace for readings, quizzes, data files, and submissions.

Grading

Weights and Points

Component	Weight	Points
Quizzes (11 given, lowest dropped; 10 × 20 pts)	20%	200
2 Small Projects (75 pts each)	15%	150
Final Project	15%	150
Exam 1	10%	100
Exam 2	15%	150
Final Exam (cumulative)	25%	250
Total	100%	1000

Grade Scale

Grade	Percentage	Points
A	93–100%	930–1000
A–	90–92%	900–929
B+	87–89%	870–899
B	83–86%	830–869
B–	80–82%	800–829
C+	77–79%	770–799
C	73–76%	730–769
C–	70–72%	700–729

Grade	Percentage	Points
D+	67–69%	670–699
D	64–66%	640–669
D–	60–63%	600–639
F	< 60%	< 599

Final percentages are **not** rounded up across grade boundaries.

Assessment Details

Quizzes (200 pts). Short quizzes on the previous sessions' material, given in class on the dates marked in the schedule. Eleven quizzes are given and the **lowest score is dropped** — this drop is how missed quizzes are handled, so there are no make-up quizzes.

Small Projects (150 pts). Two applied Excel projects. Project 1 covers data cleaning and descriptive/visual summary of a real dataset. Project 2 covers estimation and hypothesis testing. Each is submitted as a single Excel workbook plus a short written interpretation (2 pages max). The interpretation is where most of the points live: a correct number with no business meaning attached earns partial credit at best.

Final Project (150 pts). A small research project of your own design, done in teams of 3–4: pose a business question, obtain data, analyze it with the methods from this course (including regression), and report your findings. Milestones:

- **Topic proposal** (1 page, ungraded but required) — due Tue, Nov 17
- **Written report + workbook** — due Thu, Dec 10
- **In-class presentation** (8 minutes) — Thu, Dec 10

Exams. Exam 1 and Exam 2 are given in class. The Final Exam is cumulative and is given during MSU's Final Week (Dec 14–17). Exams are closed-book; a formula sheet is provided and one double-sided page of handwritten notes is permitted. Excel is used on the exams for computation.

Course Schedule

29 class meetings. Chapter numbers refer to Doane & Seward, 7e. The schedule is a plan, not a contract — changes will be announced in class and on Brightspace.

#	Date	Topic	Reading	Due / Event
1	Thu, Aug 27	Course overview; what business analytics is and is not	Ch 1	
2	Tue, Sep 1	Data types, measurement scales, sampling methods; Lab: Excel setup, enabling the Analysis ToolPak	Ch 2	
3	Thu, Sep 3	Lab: Data cleaning — tables, sorting, filtering, missing values, duplicates	Ch 2	Quiz 1
4	Tue, Sep 8	Describing data visually: histograms, scatterplots, line charts; how charts mislead	Ch 3	
5	Thu, Sep 10	Lab: PivotTables and charts in Excel	Ch 3	Quiz 2
6	Tue, Sep 15	Descriptive statistics: center, variability, shape	Ch 4	
7	Thu, Sep 17	Standardization, outliers, boxplots; Lab: Descriptive Statistics tool	Ch 4	Quiz 3 · Small Project 1 assigned
8	Tue, Sep 22	Probability: events, joint and conditional probability, independence	Ch 5	
9	Thu, Sep 24	Discrete distributions: binomial and Poisson in Excel	Ch 6	Quiz 4
10	Tue, Sep 29	Review for Exam 1	Ch 1–6	Small Project 1 due
11	Thu, Oct 1	EXAM 1 (Ch 1–6)		
12	Tue, Oct 6	The normal distribution; NORM.DIST / NORM.INV; the exponential distribution	Ch 7	
13	Thu, Oct 8	Sampling distributions and the Central Limit Theorem; Lab: simulating the CLT	Ch 8	Quiz 5
14	Tue, Oct 13	Confidence intervals for a mean (σ known and unknown); the t distribution	Ch 8	
15	Thu, Oct 15	Confidence intervals for a proportion; choosing a sample size	Ch 8	Quiz 6
16	Tue, Oct 20	Logic of hypothesis testing; one-sample tests for a mean	Ch 9	
17	Thu, Oct 22	p -values, Type I/II error, power; one-sample test for a proportion	Ch 9	Quiz 7 · Small Project 2 assigned
18	Tue, Oct 27	Two-sample tests: comparing two means	Ch 10	

#	Date	Topic	Reading	Due / Event
19	Thu, Oct 29	Paired <i>t</i> -test, two proportions, comparing variances; Lab: Excel <i>t</i> -Test tools	Ch 10	Quiz 8 · Final Project assigned
20	Tue, Nov 3	One-factor ANOVA; multiple comparisons	Ch 11	
21	Thu, Nov 5	Chi-square goodness-of-fit and tests of independence	Ch 15	Quiz 9 · Small Project 2 due
22	Tue, Nov 10	Review for Exam 2	Ch 7–11, 15	
23	Thu, Nov 12	EXAM 2 (Ch 7–11, 15)		
24	Tue, Nov 17	Statistical process control: common vs. special cause variation, control charts	Ch 17	Final Project topic proposal due
25	Thu, Nov 19	Lab: $\bar{x}$ and <i>R</i> charts, <i>p</i> -charts, process capability	Ch 17	Quiz 10
—	Nov 23–27	Fall Break — no classes		
26	Tue, Dec 1	Simple regression: least squares, r^2 , inference on the slope	Ch 12	
27	Thu, Dec 3	Multiple regression: interpreting coefficients, <i>F</i> -test, adjusted r^2 ; Lab: Excel Regression output	Ch 13	Quiz 11
28	Tue, Dec 8	Multiple regression continued: dummy variables, multicollinearity, residual diagnostics	Ch 13	
29	Thu, Dec 10	Final Project presentations; Final Exam review		Final Project report due
—	Dec 14–17	FINAL EXAM (cumulative) — day and time per the MSU Final Exam Schedule		

Final Exam time: MSU assigns final exam slots by meeting pattern. Confirm the exact date and time for a TTh 12:15 pm class on the Registrar's published Final Exam Schedule; it will also be posted to Brightspace by Week 12. University policy does not permit taking the final early.

Important University Dates — Fall 2026

Date	Event
Wed, Aug 26	Classes begin (our first meeting is Thu, Aug 27)
Tue, Sep 1	Last day to self-add
Mon, Sep 7	Labor Day — no classes (does not affect this section)
Wed, Sep 9	Last day to add via instructor; last day to self-drop
Wed, Sep 16	Last day to drop via advisor without a "W"; last day to confirm bill

Date	Event
Oct 27 – Nov 17	Priority registration for Spring 2027
Wed, Nov 11	Veterans Day — no classes (does not affect this section)
Wed, Nov 18	Last day to drop with a “W” via advisor
Nov 23–27	Fall Break — no classes
Dec 14–17	Final Week
Thu, Dec 17	Last day of finals

Neither Labor Day nor Veterans Day falls on a Tuesday or Thursday in 2026, so **this section loses no meetings to holidays** — only Fall Break.

Course Policies

Attendance and Participation

Attendance is expected. Much of this course is hands-on work in Excel that is difficult to reconstruct from slides alone. If you must miss class, get the material from a classmate and from Brightspace before coming to office hours.

Late Work

Projects lose 10% per calendar day late, up to three days; after that they are not accepted. Quizzes cannot be made up — the dropped-lowest policy exists for that purpose. If a documented emergency or a university-sanctioned activity affects a deadline or an exam, contact me **in advance** whenever possible and we will make arrangements.

Missed Exams

Make-up exams are given only for documented, unavoidable conflicts arranged in advance. An unexcused missed exam receives a zero.

Use of AI Tools

Generative AI (ChatGPT, Claude, Copilot, and similar) is permitted for *learning* — explaining a concept, decoding an error message, understanding what a statistic means. It is **not** permitted to generate your submitted analysis, your written interpretation, or your project report. On every project you must be able to explain, in your own words, every number and every sentence you submit; I may ask you to do so. If you use an AI tool on a project, include a one-paragraph note in your submission describing what you used it for. Undisclosed AI-generated work is handled as academic misconduct.

Academic Integrity

All work is governed by the MSU Student Conduct Code. Collaboration is encouraged on practice problems and required on the Final Project; Small Projects and all exams must be your own individual work. Copying another student’s workbook, submitting data you did not actually collect or obtain, or misrepresenting results are all violations.

Disability Accommodations

MSU is committed to providing equal access. If you have a documented disability and need accommodations, please register with MSU **Disability Services** (Romney Hall; 406-994-2824; drv@montana.edu) and contact me early in the semester so we can implement your accommodations. Accommodations cannot be applied retroactively to work already submitted.

Title IX and Non-Discrimination

MSU prohibits discrimination and harassment, including sexual misconduct. Instructors are required to report disclosures of sexual misconduct to the university's Title IX Coordinator. For confidential support, contact the VOICE Center (406-994-7069, available 24/7) or MSU Counseling & Psychological Services.

Student Support

- **Brightspace** — readings, data files, quizzes, grades
- **MSU Writing Center** — help with project reports
- **Counseling & Psychological Services** — 406-994-4531
- **Office hours** — the most efficient help available; bring your workbook and your specific question

Communication

Email is the official channel; I aim to respond within one business day on weekdays. Use your MSU address and include "BMGT 240IS" in the subject line. Grade questions require an in-person conversation, not email.

A Note on How This Course Works

Statistics is learned by doing it, not by watching it. Most sessions pair a concept with an Excel workbook, and you will get more out of the course if you open the file and follow along rather than watch. When a result surprises you, that surprise is the useful part — bring it to class.

Last updated: August 11, 2026. This syllabus is subject to change; changes will be announced in class and posted on Brightspace.